

2 Sea trout, *Salmo trutta*, are found in rivers and coastal seawaters in many regions of the world.

Sea trout lay their eggs in rivers and the young fish remain in this freshwater environment. After living for 1–8 years in freshwater rivers, sea trout migrate to the sea. Adult sea trout feed in coastal seawater regions, usually no more than 80 km from their home river.

Fish farms that produce Atlantic salmon, *Salmo salar*, may be located in the coastal seawater regions where wild sea trout are also found.

Fig. 2.1 shows one cage at a fish farm.

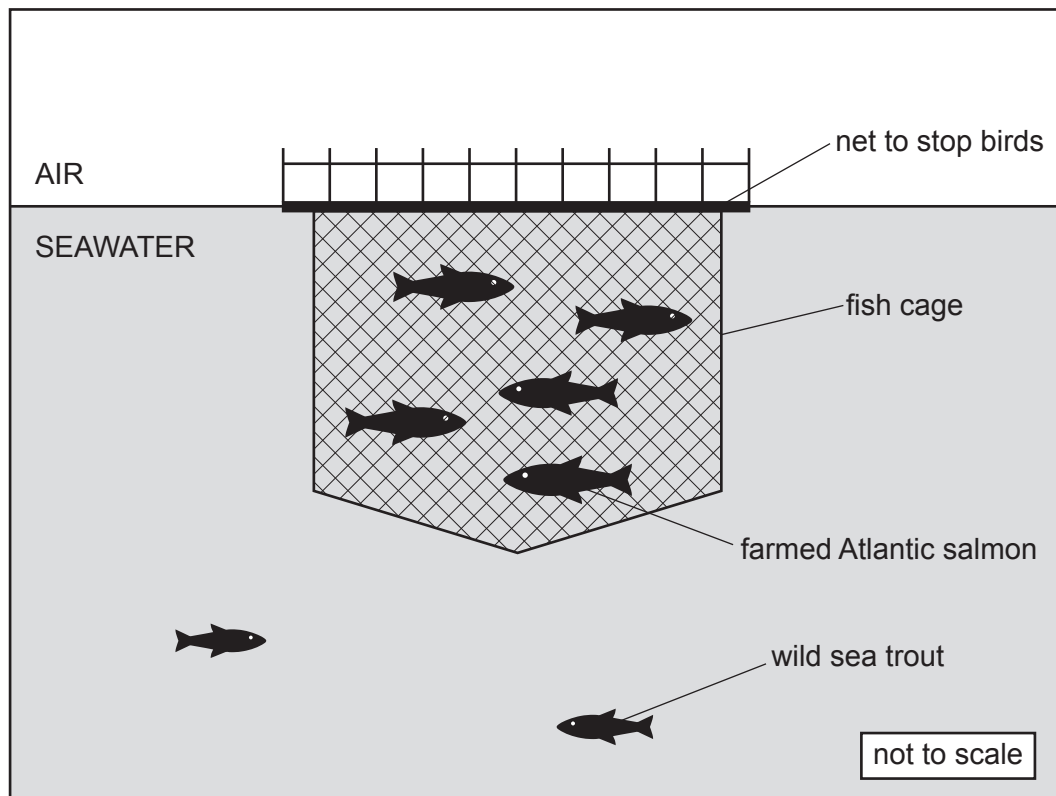


Fig. 2.1

The increase of fish farming is thought to have caused more infections by salmon lice, *Lepeophtheirus salmonis*, in wild fish such as sea trout. When sea trout enter coastal seawater regions, they can become infected with salmon lice.

Salmon lice larvae live in seawater and can move freely between fish cages and open seawater. Salmon lice attach to the skin of sea trout and feed on their skin and blood. The resulting tissue damage can sometimes cause the death of sea trout.

Fig. 2.2 shows a sea trout infected with salmon lice.

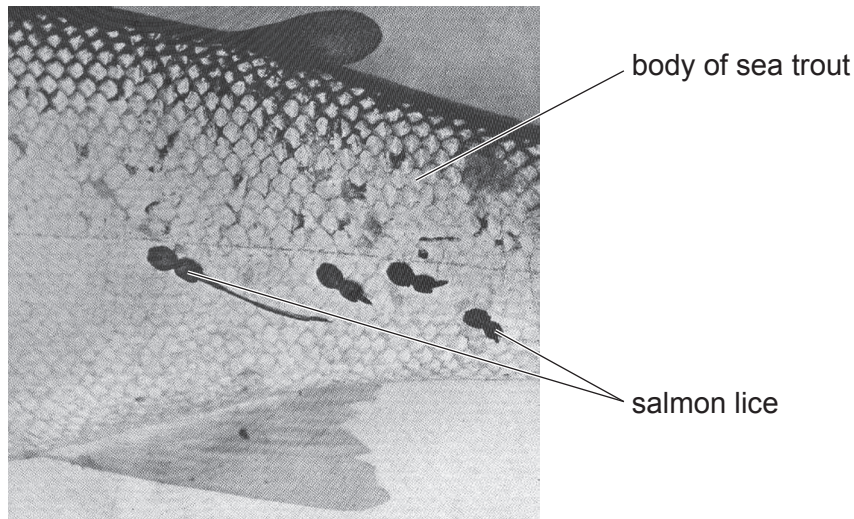


Fig. 2.2

- (a) Some scientists found out from a published source that 13 or more attached salmon lice usually cause the death of a sea trout kept in laboratory conditions.

The scientists decided that each **wild** sea trout with 13 or more salmon lice is at risk of physiological stress and potential death.

Outline a laboratory method that could be used to determine the number of salmon lice needed to cause the death of a sea trout.

Suggest a suitable control that could be used to confirm that it is the salmon lice that are causing the death of the sea trout.

method that could be used

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suitable control

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- (b) The scientists investigated the relationship between fish farming and infections by salmon lice on wild sea trout in coastal seawaters at Shielraig on the west coast of Scotland in north-west Europe.

Fish farms use a two-year production cycle for Atlantic salmon. After hatching in freshwater tanks, the young salmon are transferred to seawater cages. Atlantic salmon spend two years growing in the seawater cages before they are harvested.

Infections by salmon lice on the farmed salmon are known to be higher in the second year of the two-year production cycle.

The scientists wanted to test the hypothesis:

Infections by salmon lice on wild sea trout in coastal seawaters at Shielraig will be higher in the second year of the two-year production cycle of the fish farm near Shielraig.

Electrofishing was used to sample the wild sea trout population in coastal seawater regions near the fish farm.

- A boat was used to travel to the sample location.
- An electric current was applied to the water around the boat. This causes some fish in the area to stop moving temporarily.
- A hand net was used to catch the fish that were not able to move.
- The number of sea trout caught was recorded.
- The number of salmon lice attached to each sea trout was also recorded.
- The fish were then returned to the sea.
- Sampling was repeated once a year for 10 years.

- (i) State **one** variable that should be standardised in this procedure, **other than** the boat and the hand net.

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- (ii) Electrofishing is more likely to prevent movement in larger fish.

State why this systematic error may give results that are **not** representative of the wild sea trout population.

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- (c) The scientists collected data on infections by salmon lice on wild sea trout in coastal seawaters at Shieldaig every year from 2000 to 2009. During this time the fish farm near Shieldaig completed their two-year production cycle five times.

The data are shown in Table 2.1.

Table 2.1

year	year of production cycle	number of sea trout sampled	percentage of sea trout with attached salmon lice	percentage of sea trout with ≥ 13 attached salmon lice
2000	1	15	13	data not collected
2001	2	87	31	data not collected
2002	1	31	6	3
2003	2	44	20	14
2004	1	15	0	0
2005	2	26	0	0
2006	1	39	8	5
2007	2	30	87	70
2008	1	14	0	0
2009	2	21	33	19

- (i) Explain why the scientists calculated the percentages of sea trout with attached salmon lice.

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The scientists then calculated the mean percentage of sea trout with attached salmon lice in the first and second years of the fish farm production cycle.

These means are shown in Table 2.2.

Table 2.2

year of production cycle	mean percentage of sea trout with attached salmon lice	mean percentage of sea trout with ≥ 13 attached salmon lice
1	6.1	2.6
2	33.2	14.9

The scientists concluded that greater numbers of wild sea trout in coastal seawaters at Shieldaig are at risk of death in the second year than in the first year of the two-year production cycle of the fish farm near Shieldaig.

- (ii) Use Table 2.2 to describe the evidence that supports this conclusion.

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 [1]

- (iii) Suggest limitations of the data shown in Table 2.1 and Table 2.2 that reduce confidence in this conclusion.

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 [2]

(d) The scientists then decided to use published data to investigate the effect of distance from fish farms on infection of wild sea trout by salmon lice in north-west Europe.

- 8 locations in coastal seawater regions of Scotland and 17 locations in coastal seawater regions of Ireland were sampled over several years.
- The number of salmon lice attached to each wild sea trout sampled was recorded.
- For each location, the distance from the nearest fish farm was recorded.

The scientists predicted that the number of attached salmon lice on wild sea trout will decrease as the distance from the nearest fish farm increases.

(i) State a null hypothesis the scientists would make before carrying out statistical analysis.

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The scientists then carried out statistical tests on the data to see if the relationship between the number of attached salmon lice on wild sea trout and the distance from the nearest fish farm was significant.

The probability values (p) from the results of the statistical tests are shown in Table 2.3.

Table 2.3

location	value of p	significance
Ireland	0.001	
Scotland	0.559	

(ii) Complete Table 2.3 by indicating if the results of the statistical tests are significant **or** not significant. [1]

Fig. 2.3 shows the distance from the nearest fish farm plotted against the mean number of attached salmon lice per wild sea trout.

The error bars in Fig. 2.3 show 95% confidence intervals (95% CI).

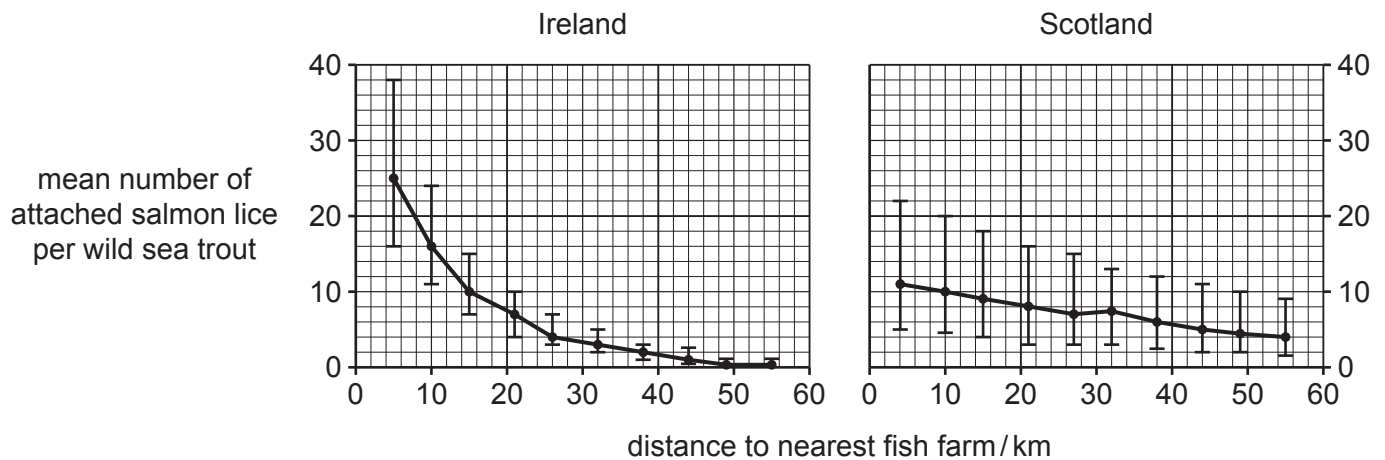


Fig. 2.3

- (iii) With reference to Fig. 2.3, state the relationship between the distance from the nearest fish farm and the mean number of attached salmon lice per wild sea trout in the coastal seawater regions of Ireland.

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 [1]

- (iv) With reference to Fig. 2.3, suggest why the scientists calculated the 95% confidence intervals for their data.

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 [2]

[Total: 15]